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Design of a Low-noise Instrumentation Amplifier with Triple-MOS Pseudo-resistor for Bio-signal Acquisition in Consumer Electronics

Feng Yan, Jingjing Liu, *Member, IEEE*, Kangkang Sun, Bingjun Xiong, Wenji Mo, Jian Guan, and Zhipeng Li

***Abstract*—Low input-referred noise (IRN) and low-power instrumentation amplifier (IA) with small size are crucial for consumer bio-electronic applications. This paper presents a compact, noise-efficient capacitive-coupled IA designed for bio-signal acquisition. The proposed Triple-MOSFET pseudo-resistor (PR) provides ultra-high resistance, a wide operating voltage range, and high linearity. The IA, enhanced by chopper-stabilized technology, effectively reduces noise while minimizing current consumption. An area-efficient DC servo loop with Triple-MOS devices is employed to suppress electrode DC offset. Addi-tionally, parasitic effects are minimized based on layout design, maximizing the advantages of the positive feedback loop. The proposed IA is fabricated using a standard 180 nm CMOS process with an area of 0.49 mm², and operates at a supply of 0.8 V with a current consumption of 1.29 µA. The IRN of this IA is 0.97 µV$_{rms}$ within the 0.5–200 Hz and 2.31 µV$_{rms}$ within the 0.2–5 kHz, achieving competitive noise efficiency factor (1.4) and power efficiency factor (1.56). Furthermore, for a 2 mVpp sine wave input, total harmonic distortion is below -60.2 dB. The precise value of the positive feedback capacitor ensures that the input impedance reaches 628 MΩ at 50 Hz. A prototype test system based on the IA successfully acquired high-quality ECG and EEG signals from subjects. This efficient, low-noise, low-power acquisition chip has broad applications in portable cardiac monitors, brain-machine interfaces, and other biosensors, playing a significant role in enhancing the performance and reliability of these electronic devices.**

***Index Terms*—Instrumentation amplifier (IA), bio-consumer electronics, low input-referred noise, MOSFET pseudo-resistor.**

I. Introduction

WITH the continuous advancement of portable device technology, low-noise measurement systems have become important tools in consumer-grade medical health monitoring [1]. These devices provide a non-invasive and comfortable way to record and monitor bioelectric signals, such as electroencephalogram (EEG), electrocardio-gram (ECG), and electromyogram (EMG) in real-time, offering users a convenient and continuous health management solution. This measurement approach can be used not only for daily health monitoring of patients but also for providing precise bioelectric characteristic data to support early disease diagnosis and personalized treatment. Due to their lightweight nature, portable measurement devices are particularly suitable for long-term wear, enabling continuous monitoring of bioelectric signals, thus offering new perspectives for research in medicine and sports health [2], [3].

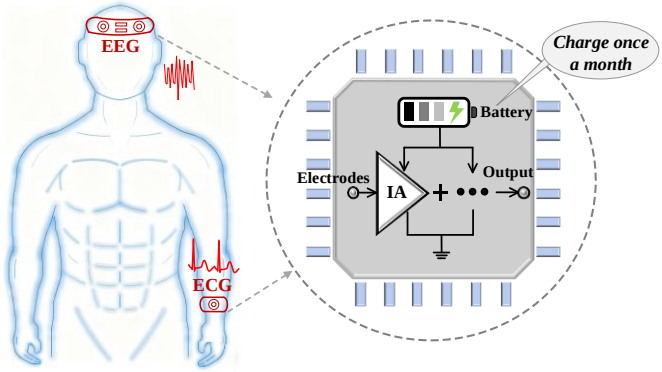

Fig. 1. The bio-consumer electronics for measuring vital signs features small size, low power consumption, and low noise.

Most existing bio-signal acquisition systems (such as ECG and EEG monitoring) still rely on bulky modular structures or dedicated medical instruments, which limits their adoption in consumer bio-electronic applications [4], [5]. To enhance user comfort, reduce system size, and promote ubiquitous health monitoring, we propose a prototype system designed for bio-consumer electronics, as illustrated in Fig. 1. The system integrates a battery and an instrumentation amplifier (IA) chip, where bio-potentials are sensed through skin-contact electrodes and transmitted via connecting wires to the IA for amplification and processing. The IA in portable biosensors is a critical component in the signal chain, playing a decisive role in high-precision bioelectric measurements. Since bioelectric signals are extremely weak and highly susceptible to noise and environmental interference [6], the IA must accurately amplify these signals while minimizing distortion to ensure reliable acquisition. In biomedical applications, a high input impedance is required to prevent tissue damage caused by electrode DC currents [7] and to minimize signal attenuation. As the IA directly influences the sensitivity and accuracy of the measurement system, its performance often dictates the overall system effectiveness [8]. Therefore, optimizing IA design is

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essential to enhance the precision and robustness of portable biosensing systems.

To achieve long-term monitoring in bio-consumer electronics, the IA needs not only to have high-precision amplification capabilities but also to exhibit low power consumption and low noise characteristics [9]. Low supply design is intended to extend the system's operational life, especially when battery capacity is limited [10]. For example, a small sized 620 mAh zinc-air button signal measurements, excessive noise can distort the signal and cause the system to fail. Accurate acquisition of bioelectric signals requires the IA chip to have input reference noise (IRN) levels as low as the microvolt range. Since there is a trade-off between noise and power consumption, the noise efficiency factor (NEF) [11] and power efficiency factor (PEF) [12] are commonly used as performance metrics. Lower NEF and PEF values indicate that the amplifier achieves higher power efficiency while maintaining low noise, reflecting its potential for high-precision biosensor applications.

This paper demonstrates a low-power chopper IA that achieves an optimal NEF of 1.4, providing a solution for precise signal acquisition in biosensors. The proposed IA prototype, based on a high linearity triple-MOS pseudo-resistor (PR), is implemented in 180 nm CMOS process and can be easily integrated into common portable devices. This IA chip achieves an IRN of 0.97 μV_{rms} at 0.8 V voltage. Clear EEG and ECG measurement results demonstrate that the proposed IA not only effectively reduces system power consumption but also enables high-precision bio-signal acquisition under low-noise conditions. The remainder of this paper is organized as follows: Section II provides a brief review of the progress in the NEF of IAs. Section III presents the architecture and the noise analysis of the proposed IA. Section IV describes the detailed performance results of the IA chip. Section V demonstrates human body measurement results obtained from a detection system built with the proposed IA, comparing these results with state-of-the-art technologies. Finally, Section VI concludes the paper.

II. Overview of IA Chips

The IA serves as a fundamental building block in biosensing systems, and its architectural design critically influences the overall noise performance, power efficiency, and level of integration [13], [14]. As shown in Fig. 2, the traditional three-op-amp architecture typically consists of simple two-stage amplifiers with gain set by external resistors, offering high input impedance and good common-mode rejection ratio (CMRR). However, this architecture suffers from high power consumption due to the resistive load, requiring substantial power to maintain performance. The current-feedback IA [15] reduces power consumption by sharing the output stage while maintaining high input impedance, but its gain is affected by transconductance mismatch. The current-balanced IA [16] is not influenced by this and can enhance performance through appropriate power adjustment, making it suitable for systems with fewer channels. The capacitive-coupling IA architecture [17], a commonly used topology, employs operational amplifiers (OTAs) and passive resistor-capacitor networks. This configuration provides high input impedance with low power consumption and suppresses electrode DC offset [18], but it requires careful selection of low-noise and stable resistors and capacitors to avoid introducing new noise and errors.

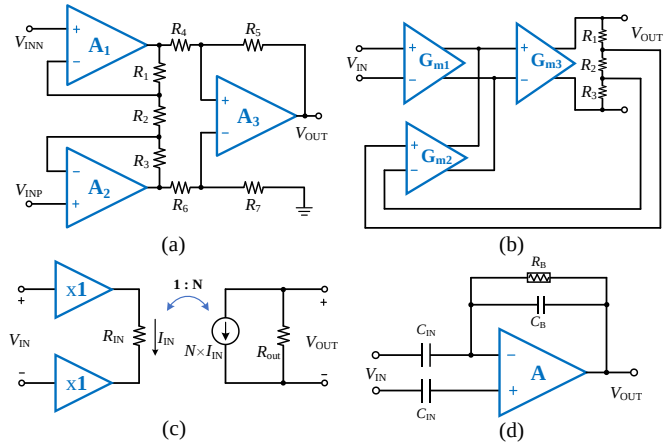

Fig. 2. Schematic diagram of IA architecture: (a) traditional three-op-amp, (b) current-feedback, (c) current-balanced, (d) capacitive-coupling.

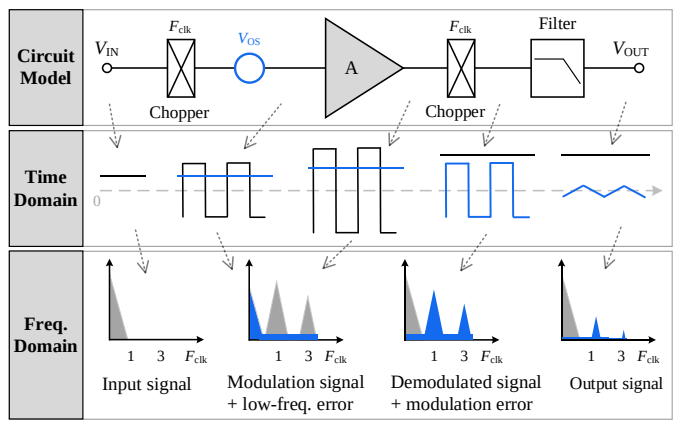

Fig. 3. The circuit model of IA with chopper stabilization technique and variations in time- and frequency-domains.

Usually, noise efficiency is denoted by NEF, which is defined as (1).

$$NEF = v_{ni,\,rms}\sqrt{\frac{2\cdot I_{tot}}{\pi\cdot 4kT\cdot V_T\cdot BW}} \tag{1}$$

where, $v_{ni,rms}$ is the total IRN of IA chip, I_{tot} is the supply current, V_T is the thermal voltage at absolute temperature T and BW is the −3 dB bandwidth of the IA. The PEF defined as $NEF^2 \times$VDD captures the actual noise-power tradeoff with VDD as the supply voltage.

Various techniques have been reported to reduce the NEF and PEF of amplifiers, such as complementary input pairs, current reuse, and OTA sharing. An NEF of 4 is achieved using the design methods described in [12], approaching the theoretical lower bound of 2.9 determined by the selected OTA architec-ture. The work in [19] employs a partial OTA sharing structure, achieving an NEF of 3.35—below the theoretical of traditional differential pair-based amplifiers. In [20], NEF was reduced to 3.09 by adopting current scaling and current splitting. OTAs based on very-low transconductance can achieve a NEF of 2.37 with a 1.8 µA current [21]. Although these techniques have made progress in improving performance, they still face challenges related to flicker noise, which severely affects the signal quality. The introduction of chopper-stabilized techniques is employed to suppress amplifier flicker noise by shifting the 1/f frequency corner to

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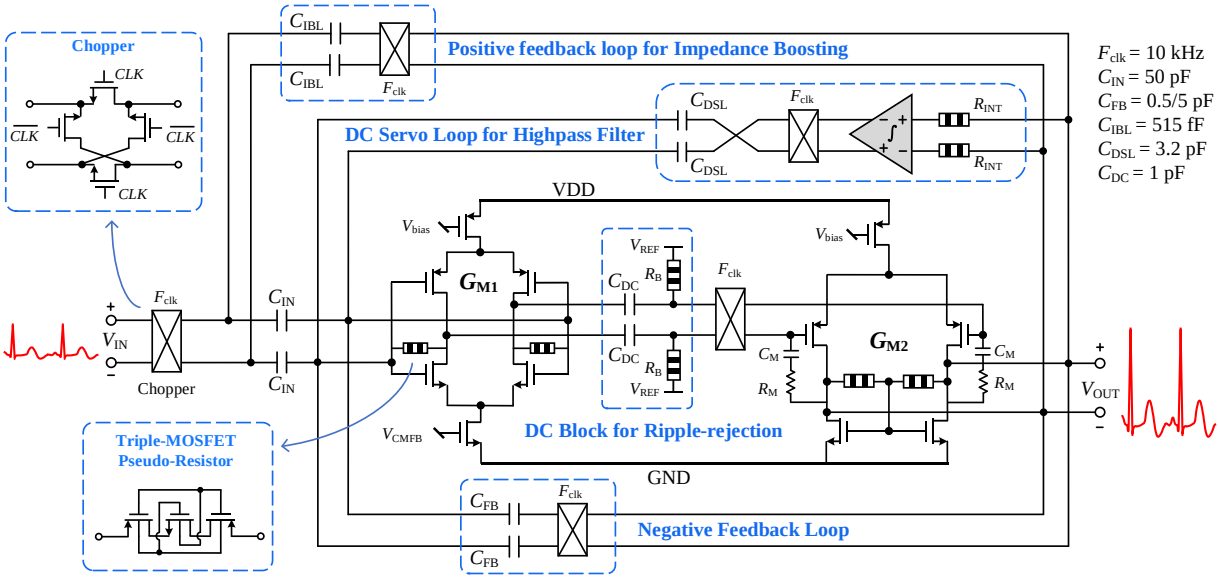

Fig. 4. Architecture of the proposed chopper-stabilized instrumentation amplifier for bio-electrical signal acquisition.

sub-Hz, achieving an IRN lower than 1 μV_{rms} while maintaining a low NEF.

Chopper stabilization suppresses low-frequency noise (e.g., 1/f noise and electrode DC offset) by modulating the input signal with a square wave [22]. As shown in Fig. 3, this modulation shifts low-frequency errors out of the signal band, thereby improving low-frequency performance. The flicker noise is translated to higher frequencies, significantly reducing the baseband noise power spectral density and the input-referred noise. This enables low-noise operation at reduced bias current while maintaining sufficient gain–bandwidth product. Consequently, low noise and low offset are achieved, with improved CMRR and power efficiency. A DC servo loop (DSL) further compensates electrode DC offset via integration feedback, preserving the signal-chain dynamic range [23]. Chopper stabilization significantly reduces NEF without increasing power consumption, making it well suited for high-input-impedance applications. Moreover, operating the chopper-stabilized AFE at a reduced supply voltage directly contributes to improved power efficiency, enabling a lower PEF while maintaining the required noise performance for bio-signal acquisition.

III. Design of Low-Noise And Low-Power IA

A. System Overview of the Proposed IA

The proposed IA is designed based on chopper-stabilized technology to reduce noise and non-ideal factors. Fig. 4 show the circuit architecture of the IA chip with Triple-MOS PR. It consists of: 1) a two-stage Miller-compensated OTA; labeled as G_{M1} and G_{M2}; 2) a positive feedback loop for input impedance boosting; 3) a DC block for ripple-rejection; 4) an integrator-based DSL using the proposed Triple-MOS PR; and 5) a capacitive negative feedback loop. The chopper is realized by the minimum size NMOS. The mid-band gain of the IA depends on the ratio of the input capacitance C_{IN} to the feedback capacitance C_{FB}. In this work, the mid-band gain is designed to be tunable between 20 dB and 40 dB, allowing bioelectric signals to be amplified to the millivolt level to meet the requirements of subsequent system stages. Considering the trade-offs among chip area, noise performance, and matching accuracy, the input capacitor C_{IN} is selected as 50 pF. This choice provides improved noise immunity and robust gain accuracy, resulting in a feedback capacitance C_{FB} of 0.5/5 pF.

Two-stage OTAs are selected to provide an open-loop gain greater than 100 dB, ensuring the IA's closed-loop accuracy exceeds 0.1 %. An input stage with a complementary structure doubles the effective transconductance without increasing the bias current. The offset and flicker noise of G_{M1} are modulated by a chopper and then filtered by the integrator constructed from G_{M2}. Since the noise of the G_{M2} is much smaller than the G_{M1}, it is possible to bias the G_{M2} to a relatively low current for better power efficiency. In G_{M2}, the Miller compensation is implemented to ensure closed-loop stability. Meanwhile, the CM feedback employs a resistive sensing scheme to improve the linearity further. The frequency of the chopper F_{clk} is typically greater than the flicker noise corner frequency of G_{M1}. For the low-noise design, the input transistors are biased in the deep subthreshold region. The current-noise PSD of the subthreshold MOSFET can be modeled as (2).

$$\overline{i_{ni}^2} = 4kT\frac{1}{2s}g_m \tag{2}$$

where g_m is the transconductance of the input transistor and s is a process-dependent subthreshold slope factor. When the noise contribution from the second stage can be neglected, the IRN PSD of the two-stage OTA can be expressed as (3).

$$\overline{v_{ni}^2} = 4\frac{\overline{i_{ni}^2}}{(2g_m)^2} = 2kT\frac{V_T}{s^2 I_D} \tag{3}$$

According to (3), the complementary input structure can reduce the IRN by a factor of √2 compared with the traditional design without such a structure. A theoretical IRN of 41 nV/√Hz is expected when the bias current of G_{M1} is set to 0.5 μA. To minimize the gain caused by the offset and flicker noise of G_{M1}, a Triple-MOS PR is placed in the feedback at both ends of the first stage. It is located in the DC feedback loop to establish a stable DC bias for the input pair.

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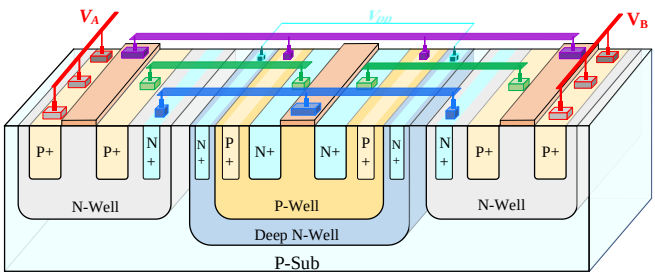


Fig. 5. The three-dimensional structure of the proposed Triple-MOS PR.

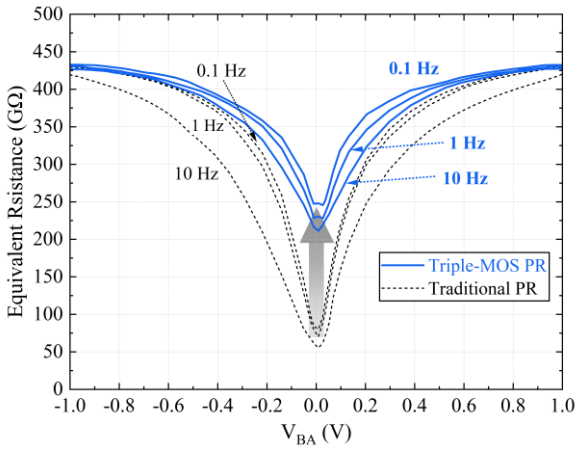
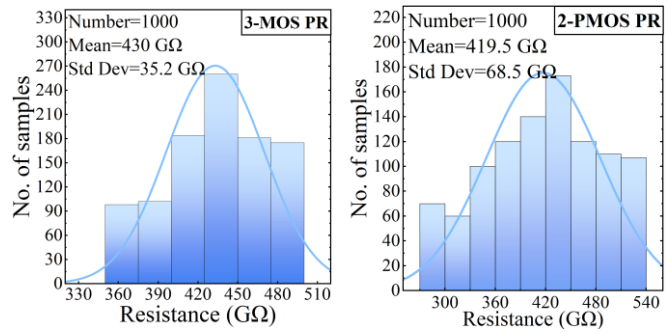


Fig. 6. Comparison of the frequency characteristics of the traditional 2-PMOS PR and the proposed Triple-MOS PR. The results show that the resistance fluctuation of the traditional PR exceeds 350 GΩ (approximately 72 GΩ at 1 Hz), whereas that of the Triple-MOS PR is reduced to about 195 GΩ.

B. Triple-MOS PR with High Linearity

The circuit structure based on high-equivalent-resistance transistors exhibits resistor-like electronic characteristics, commonly referred to as PR, and occupies much less area than physical resistors [24]. However, the PR is not an ideal element and its resistance varies with the terminal voltage, i.e., it exhibits some degree of nonlinearity. The high conduction characteristics of traditional 2-PMOS PRs at lower voltage differences cause the overall resistance to fluctuate dramatically, resulting in IA output swing compression and signal distortion. To address this drawback, a Triple-MOS PR structure is proposed. By inserting an NMOS transistor between the pair of PMOS devices, the conduction characteristics across different voltages are better balanced, thereby improving the overall linearity. The three-dimensional structure of Triple-MOS PR is shown in Fig. 5, where the gate of the NMOS transistors is connected to the substrates of the two PMOS. By embedding the NMOS into the symmetric structure, not only is the linearity enhanced, but a high resistance value is also ensured. Notably, despite the addition of extra MOSFET, this structure retains the symmetry of traditional PR. The current is proportional to the NMOS channel width, and the resistance value of the PR increases as the transistor W/L ratio decreases.

Bioelectrical signals convey information through frequency and amplitude, and consideration of the frequency characteristics of PR is critical to ensure accurate amplification of low-frequency signals in IAs. Fig. 6 shows the resistance characteristics of the Triple-MOS PR at different frequencies. The results indicate that near V_{BA}=0 V, the resistance values of the



(a) (b)

Fig. 7. Results of 1000 Monte Carlo analysis at V_{BA}= ±1 V: (a) proposed Triple-MOS PR, (b) traditional 2-PMOS PR. The Triple-MOS PR achieves a standard deviation approximately half that of the 2-PMOS PR, confirming its enhanced linearity.

PR at all frequencies exhibit significant improvement. At 1 Hz, the minimum resistance value of the Triple-MOS PR is approximately 235 GΩ, reducing the fluctuation by about 153 GΩ compared to the traditional back-to-back 2-PMOS PR. The use of the Triple-MOS PR structure improves linearity over a wide voltage range and effectively suppresses even-order harmonics in the sinewave signal excitation at V_{BA} = 0 V.

As show in Fig. 7, the robustness of the Triple-MOS PR is verified through 1000 Monte Carlo analysis at V_{BA} = ±1 V. The results show that the average resistance of the Triple-MOS PR is 430 GΩ, while the average resistance of the back-to-back 2-PMOS PR is 419.5 GΩ. The embedded NMOS structure significantly reduces the standard deviation of its resistance value, approximately half of that of the traditional structure, indicating a substantial reduction in sensitivity to process variations. In terms of layout design, the use of cocentroid geometry and dummy structures matching the same environment effectively reduces mismatch effects. The proposed Triple-MOS PR demonstrates superior resistance stability over a wide voltage range compared to the traditional 2-PMOS PR.

C. DC Servo Loop with Triple-MOS PR

The DSL techniques suppress large electrode offsets through a negative feedback loop and attenuates the low-frequency signal at the V_{OUT}. The DSL includes an integrator which amplifies the DC signal at the output of G_{M2}; a chopper which up-modulates the amplified DC signal; and capacitors C_{DSL} which feed the up-modulated signal to the virtual ground of G_{M1}. The integrator continues integrating until the signal at the G_{M2} output no longer exhibits a DC component. The DSL creates a high pass corner f_{hp}, given by (4).

$$f_{hp} = \frac{C_{DSL}}{C_{FB}} f_{0DSL} \tag{4}$$

where f_{0DSL} is the unity gain frequency of the integrator. Thus, the differential electrode offset V_{eo} that can be attenuated after DSL is given by (5).

$$V_{eo} = \frac{C_{DSL}}{C_{IN}} V_{out\max} \tag{5}$$

where V_{outmax} is the maximum output voltage of the DSL integrator, i.e., the supply voltage. The capacitor C_{DSL} is kept relatively small to reduce in-band noise contributions from the servo-loop integrator and G_{M1}. Simultaneously, the minimum value of C_{DSL} is limited by the maximum electrode offset that

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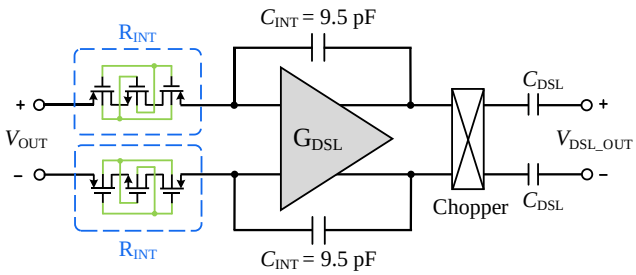

Fig. 8. Schematic of the DSL circuit based on the Triple-MOS PR, where the integration capacitor is reduced to only 9.5 pF.

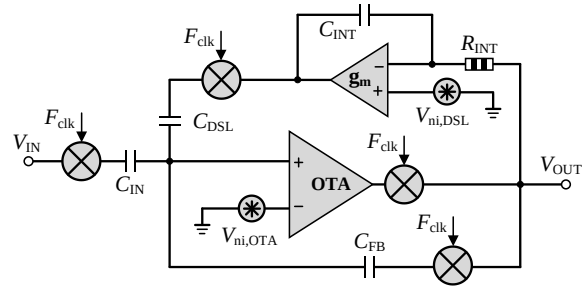

Fig. 9. Schematic of the proposed amplifier for noise analysis.

needs to be attenuated. As a trade-off between area consumption and noise, the C_{DSL} is set to 3.2 pF, which is sufficient to attenuate differential electrode offsets down to 50 mV. The f_{hp} is set to less than 0.5 Hz to meet ECG/EEG specifications; therefore, f_{0DSL} is calculated at 0.078 Hz by (4). To realize such a low corner frequency within a reasonable chip area, an integrator based on the Triple-MOS PR is implemented in the proposed IA. A block diagram is shown in Fig. 8. Theoretically, the implementation of the 430 GΩ PR reduces the required integral capacitance C_{INT} to 4.75 pF; even at the worst resistor value, the required C_{INT} is only 9.27 pF. Therefore, the C_{INT} is set to 9.5 pF. The large resistance of the high linearity Triple-MOS PR effectively reduces the required size of the integrating capacitor (C_{INT} = 40 pF in [21], C_{INT} = 20 pF in [25]). More importantly, the improved Triple-MOS PR has better linearity than the conventional PR can effectively reduce the frequency response of the integrator to shift, thus solving the phenomenon of low-frequency gain distortion or servo incomplete compen- sation. Moreover, the OTA G_{DSL} used in DSL features an AB- class structure to boost driving performance, with a total current consumption of 0.4 μA in the presence of a CM feedback loop.

D. Positive Feedback Loop and Ripple-rejection Loop

Without invoking any impedance boosting techniques, the equivalent input impedance of the chopper IA with a 10 kHz chopping frequency is roughly 1 MΩ. This is far from sufficient for a bioelectric signal acquisition chip. In traditional positive feedback techniques, the positive feedback capacitor C_{IBL} is usually set to the same size as the negative feedback capacitor C_{FB} to compensate for the current drawn by the chopper from the signal source. For instance, in [26], C_{IBL} = C_{FB} = 120 fF, enhancing the input impedance to 30 MΩ. However, this approach merely clones the positive feedback capacitance value without fully considering the parasitic capacitance effects. To further enhance the input impedance, this work optimizes the positive feedback capacitance value based on the post-layout simulation results of the IA chip. By precisely compensating for the parasitic capacitance introduced by the layout, the final capacitance value of C_{IBL} is set to 515 fF. This improvement effectively increases the input impedance of the IA and minimizes signal attenuation.

The error modulated to high frequencies by the chopper significantly reduce the available linear swing at the amplifier output, thereby limiting the dynamic range of the measurement. The proposed IA employs a low-power ripple suppression technique based on capacitive-resistive elements. To block offset current and noise that may propagate to G_{M2}, a pair of capacitors, C_{DC}, is added at the output of G_{M1}, and the DC level of G_{M2} is biased to the reference voltage via a resistor R_F. The inserted DC block effectively attenuates the chopper ripple at the output without adding extra power consumption. There are no DC current flows from G_{M1} to G_{M2}, resulting in the expected attenuation of the ripple caused by G_{M1} is theoretically infinite. The unit gain frequency F_{ug} of this DC block is given by (6).

$$F_{ug} = \sqrt{\frac{A_0 \omega_{clk} C_M}{R_F C_{DC}\left(C_{IN} + C_{FB} + C_{DSL}\right)}} \tag{6}$$

Typically, F_{ug} is greater than the flicker noise corner frequency of G_{M1}. A large bias resistor can effectively reduce the in-band noise contribution [25]. Therefore, R_F is implemented using a Triple-MOS PR with a large resistance value and high linearity, which produces an input noise density of approximately 11 nV/√Hz. This noise density is negligible compared to the overall noise density of the detection system. Meanwhile, capacitor C_{DC} is chosen to be C_M/8 to minimize the chip area.

E. Noise Analysis

A noise equivalent model as shown in Fig. 9 was built to analyze the noise performance. The IRN PSD of the proposed chopper-stabilized IA is calculated by dividing the amplifier's output noise PSD by its mid-band gain, as shown in (7).

$$\begin{aligned}\overline{v_{n,tot}^2} &= \overline{v_{ni,OTA}^2}\left(\frac{C_{IN}+C_{FB}+C_{DSL}}{C_{IN}}\frac{1}{1+\frac{2\pi f_{hp}}{s}}\right)^2 \\ &+ \overline{v_{ni,DSL}^2}\left(\frac{C_{DSL}}{C_{IN}}\frac{1}{1+\frac{2\pi f_{hp}}{s}}\right)^2 + \overline{v_{ni,ch}^2}\left[\left(\frac{C_{FB}}{C_{IN}}\right)^2+1\right]\end{aligned} \tag{7}$$

where $\overline{v_{ni,OTA}^2}$, $\overline{v_{ni,DSL}^2}$, and $\overline{v_{ni,ch}^2}$ are the PSDs of the core OTA, DSL, and chopper, respectively. For low-frequency bioelectric signals, the chopper noise can be ignored [21]. The total input-referred integrated noise from the high-pass cutoff frequency f_{hp} to the low-pass cutoff frequency f_{lp} can be calculated as (8).

$$\begin{aligned}V_{ni,rms} &= \sqrt{\int_{fhp}^{flp}\overline{v_{n,in}^2}df} \\ &= \sqrt{2kT\frac{V_T}{s^2 I_D}\left(\frac{C_{IN}+C_{FB}+C_{DSL}}{C_{IN}}\right)^2 f_{lp} + \frac{kTV_{eo}}{sGV_{out\max}C_{INT}}}\end{aligned} \tag{8}$$

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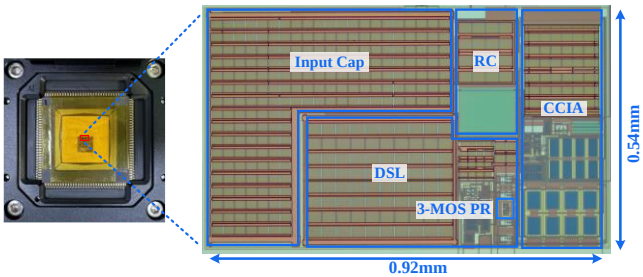

Fig. 10. The packaged chip and microphotography of the proposed IA.

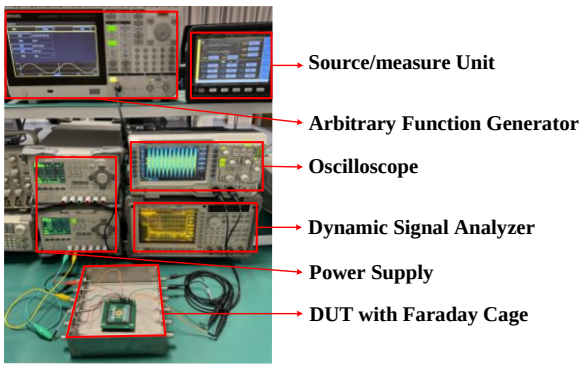

Fig. 11. The measurement platform for the proposed IA chip.

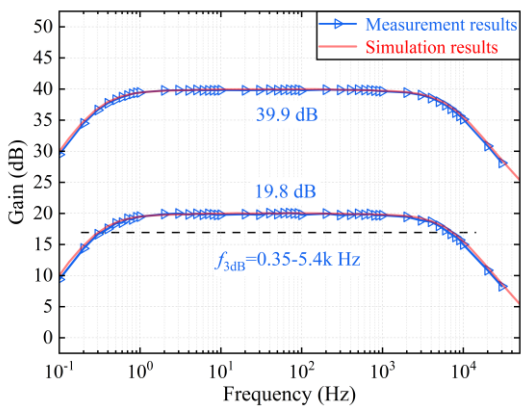

Fig. 12. Measured mid-band gain of the proposed IA, showing two adjustable gain settings of 19.8 dB and 39.9 dB.

For a typical EEG signal with a frequency range of 0.5 to 200 Hz, the IRN of the proposed IA is about 0.9 μV_{rms} according to the post-layout simulation results.

IV. Experimental Results of The IA Chip

The proposed chopper-stabilized IA chip is fabricated using a standard 180 nm CMOS process. It requires no special fabrication materials and facilitates integration into biomedical devices. The chip microphotography is shown in Fig. 10. With the dummy technique implemented for optimal matching, the effective chip area of the proposed IA is 0.49 mm^2. It consumes 1.032 µW under a 0.8 V supply. Fig. 11 shows the test bench setup for IA chip. The Keysight precision source/measure unit (B2910BL) provides the necessary current and voltage. Time- and frequency-domain analysis is performed using a dynamic signal analyzer (35670A). An active balun, composed of auxiliary OTAs, was used to realize fully differential

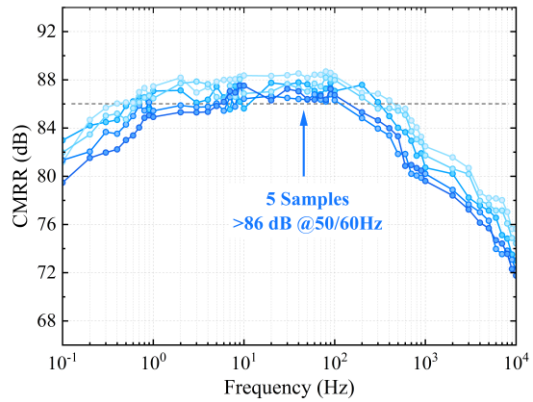

Fig. 13. Measured CMRR results of five fabricated IA.

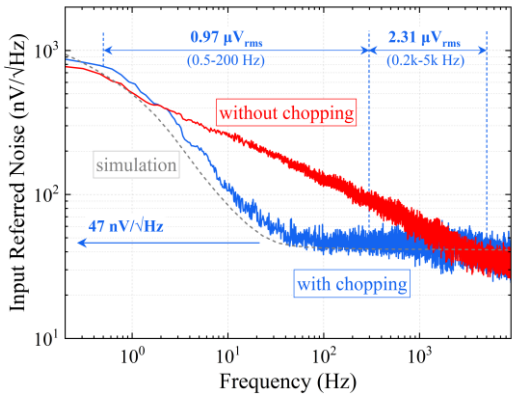

Fig. 14. Measured IRN of the IA with and without chopping for comparison.

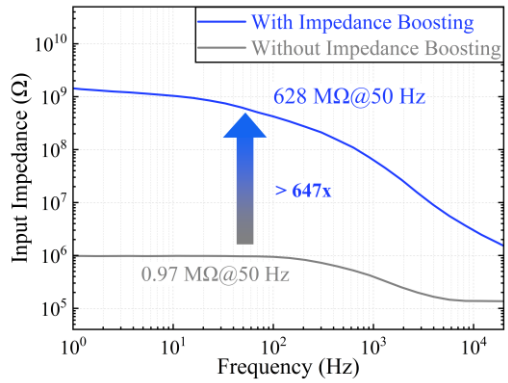

Fig. 15. Comparison of input impedance measurement results for IA.

measurements. The device under test (DUT) is placed inside a Faraday cage to minimize interference.

The measurement result of the frequency characteristics are shown in Fig. 12. The mid-band gain is 19.8 and 39.9 dB with a high-pass cutoff frequency of 0.35 Hz and a low-pass cutoff frequency of 5.4 kHz. The measurement results are in good agreement with the post-layout simulation results. The proposed IA features a wide frequency response range, enabling it to effectively capture and amplify various biosignals. Biosensors is affected by common-mode interference (CMI). The chopper technique allows for the attainment of a high CMRR. The CMRR measurement results of 5 samples at a CMI of 200 mV_{pp} are also given in Fig. 13. It can be observed that the proposed IA chip achieves a CMRR exceeding 86 dB at 50/60 Hz

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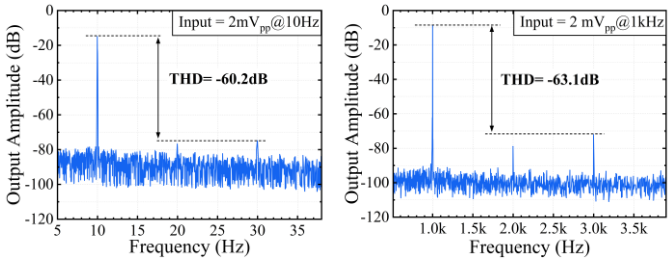

Fig. 16. The measured harmonic distortion of the amplifier, (a) for a 2 mVpp input at 10 Hz, (b) for a 2 mVpp input at 1 kHz.

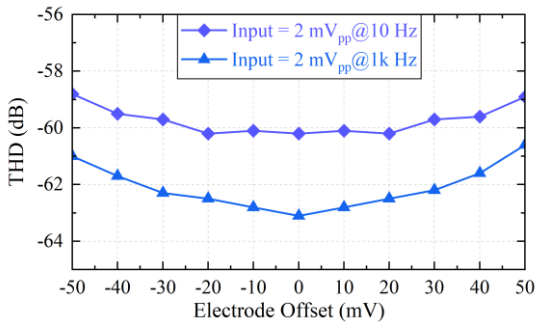

Fig. 17. The measured results of THD versus 50 mV electrode offset for a 2 mVpp input at 10 Hz and 1 kHz, respectively.

without any tuning. The drop in CMRR at high frequency band is attributed to the limited bandwidth of the amplifier.

Fig. 14 shows the IRN results of the IA chip. The noise with chopper modulation is significantly lower than that without chopper. The measured results with the chopper show good match with the post-layout simulation results. The chopper technique effectively reduces low-frequency noise, and the input reference thermal noise is 47 nV/√Hz. The measured integrated noise is 0.97 and 2.31 μV_{rms} in the frequency band of 0.5–200 Hz and 0.2–5 kHz, respectively.

The proposed IA's input impedance is indirectly obtained by quantifying the ADM attenuation induced by a known 5 GΩ input resistor [9]. Fig. 15 shows the input impedance without impedance boosting: 0.97 MΩ at 50 Hz, and the increased input impedance achieved by enabling the impedance boosting loop: 628 MΩ at 50 Hz. This value is greater than the minimum requirement of 10 MΩ for electrode measurements [27] and the impedance enhancement factor exceeds 647.

Fig. 16 presents the measurement results of total harmonic distortion (THD) at different frequencies. In the DC to 200 Hz range, the THD is dominated by the DSL, while in the 0.2–5 kHz range, the two-stage core OTA predominates. Thanks to the high linearity of the proposed Triple-MOS PR, the measured THD of the IA chip is lower than -60.2 dB. Fig. 17 illustrates the relationship between THD and electrode offset with a 2 mVpp sine input at 10 Hz and 1 kHz. Although the THD increases slightly with electrode offset, it remains below -59 dB at 50 mV.

V. Biologic Measurement

The proposed low-noise IA is validated measuring an adult ECG and EEG signals in lab environment with commercial Ag/AgCl wet electrodes. The test platform, as shown in

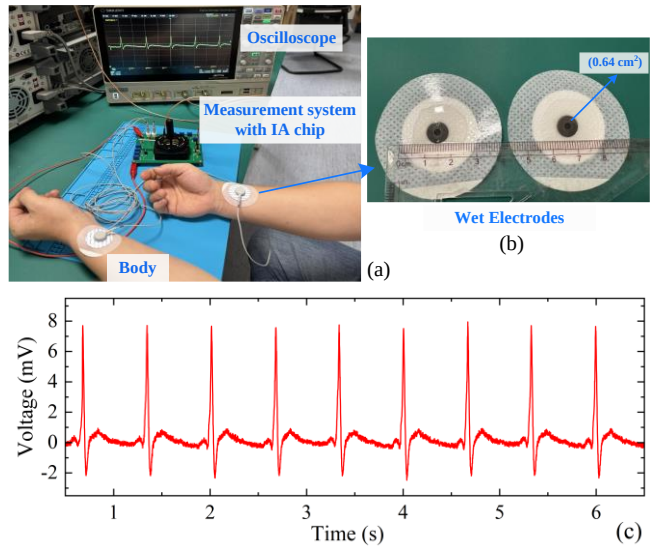

Fig. 18. (a) Biologic measurement platform and (b) test electrodes. (c) The measurement results of time-domain ECG by the IA chip.

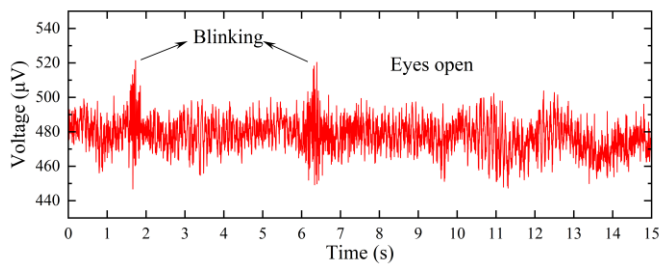

Fig. 19. EEG measurements using ear-clip electrodes.

Fig. 18(a), was built on a printed circuit board and includes only a power module and the IA chip packaged in a Quad Flat Package. The input of the detection system was connected to the test subject using 0.64 cm² electrodes, as illustrated in Fig. 18(b), while the output was routed to an oscilloscope via coaxial cable for direct waveform observation. For a more challenging test scenario, electrodes were placed on the arm instead of the chest to capture weaker ECG signals. It is noted that the subject remained at rest during the test. Fig. 18(c) presents the time-domain ECG signals acquired using the standard Lead_I configuration. The raw waveform clearly exhibits the P-wave, T-wave, and QRS complex.

For EEG acquisition, ear-clip electrodes were positioned at FP2, A2, and A1 sites [23]. The waveform activity is presented in Fig. 19. The detected variations between eye-blinking and eye-open conditions reveal a strong alpha wave rhythm, demonstrating the system's effectiveness in capturing EEG signals.

Table I summarizes the proposed IA chip performance and compares it with other IAs used for bio-consumer electronics. While some previous works [28], [29], and [17] have achieved excellent CMRR, they suffer from noise efficiency issues and high-power consumption. The IAs in [21] and [31] feature small area and low IRN, but their high supply voltage results in degraded PEF. Owing to the careful consideration of Triple-MOS PR, the proposed IA exhibits low IRN across a wide frequency range, achieving an optimal NEF of 1.4 and PEF of 1.56. The optimized positive feedback achieves comparable input impedance while maintaining low power consumption

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TABLE I
PERFORMANCE COMPARISON WITH THE STATE-OF-THE-ART IAS

Parameter	[28] 2018	[21] 2019	[29] 2020	[17] 2021	[30] 2021	[31] 2022	[3] 2023	[12] 2024	This Work	
Application	EEG	LFP/AP	EEG/ECG	EEG/ECG	EMG/ECG	LPF	IECG	ECG	EEG/ECG	
Process (nm)	65	180	130	180	180	180	180	180	180	
Supply (V)	1.2	1.8	2	1.2	1.8	1.8	0.9	1.2	**0.8**	
Power (μW)	5.4	3.24	6.3	2.76	43.8	8.3	4.5	1.1	**1.032**	
Gain (dB)	40	40	43–55	40	35	51	26–44	49.2	39.9	
Area (mm^2)	N/A	0.2/ch	0.6	1.57/ch	0.75	0.026/ch	0.09	0.011	0.49	
THD (dB)	N/A	-61 @5mV_{pp}	-68 @6mV_{pp}	N/A	N/A	57	60 @10mV_{pp}	40 @1kHz	-60.2 @2mV_{pp}	
Impedance(GΩ)	1	0.44	N/A	N/A	0.16 @50Hz	1.5	N/A	N/A	0.628 @50Hz	
CMRR (dB)	110	100	112	110	76	117	75	40	86	
PSRR (dB)	N/A	70	101	83	N/A	78	68	N/A	81	
IRN (μV_{rms}@Hz)	0.44 (0.5–100)	0.65 (0.3–200)	2.4 (1–40)	3.2 (0.5–400)	3.52 (1–150)	0.96 (1–200)	2.7 (0.15–200)	12.7 (0.21–8.7)	**(0.5–200) 0.97**	**(0.2–5k) 2.31**
NEF	3.59	2.37	13.2	9.4	54.6	5.62	2.92	4.92	**2.9**	**1.4**
PEF	15.46	10.11	52.8	106	5366	56.85	7.67	29	**6.72**	**1.56**

LPF: local field potential; AP: action potential.

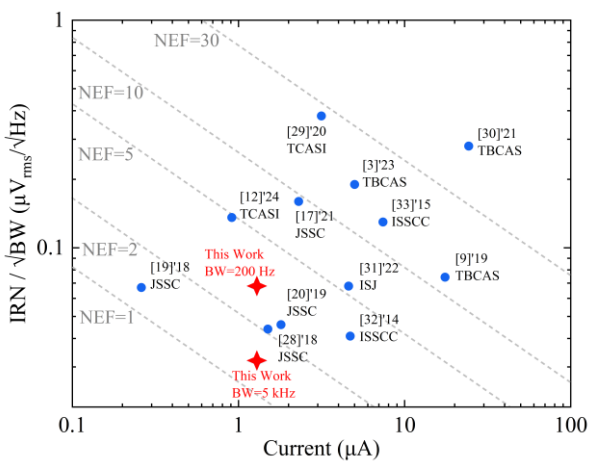

Fig. 20. The comparison between the proposed IA and the state-of-the-art IAs in terms of input-referred RMS noise versus supply current.

and small area. Fig. 20 shows the relationship between IRN and the current consumed by the amplifier. Compared to state-of-the-art works, the proposed IA chip demonstrates lower noise while achieving a competitive NEF.

VI. CONCLUSION

In this paper, a low-noise and low-power IA with chopper modulation topology is proposed for bioelectric signal acquisition. The operation principle, circuit implementation, noise analysis, and measurement results of the proposed IA have been described in detail. The improved Triple-MOS PR comprehensively enhance the performance of the IA chip. The DSL suppresses the electrode offset to ±50 mV with an integration capacitance of only 9.5 pF. A simple area-efficient ripple suppression technique is implemented to drastically reduce the chopping ripple. Meanwhile, a positive feedback loop that eliminates parasitic effects according to the layout increases the input impedance by a factor of 647. The proposed IA occupies 0.92 × 0.54 mm, consumes 1.032 μW, and achieves > 86 dB CMRR at 5 samples without any off-chip tuning. The IA chip with Triple-MOS PR achieves an INR of 0.97 μV_{rms} at 0.5–200 Hz, and realizes a NEF of 2.9, ensuring the comparable input impedance and the best noise performance. The proposed IA has been validated for biologic measurements and provides a high-performance, low-cost solution for consumer-grade electronics.

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